Yonghan Jiao

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Technical Skills

Simulation COMSOL Multiphysics, ANSYS Fluent, structured meshing, user-defined functions

Programming Python, MATLAB, C++, Julia

Testing and analysis Polarization, vapor collection, battery pack testing, EIS, cyclic voltammetry

Education

Carnegie Mellon University 2025–Present

M.S. in Mechanical Engineering, Research Track | GPA 3.96/4.00

Xi'an Jiaotong University 2021–2025

B.Eng. in Energy Storage Science and Engineering, Qian Xuesen Honors College

Research Experience

Flowless Zinc–Bromine Batteries | Carnegie Mellon University Jun 2026–Present

- Developed a transient COMSOL model coupling electrochemical kinetics, species transport, natural convection, and bromine storage in a membraneless cell.
- Compared modeled voltages and resistances with published measurements and began developing an electrode scaffold to test spatial predictions in situ.

Regenerative Fuel Cells | Xi'an Jiaotong University May 2026–Present

- Built 3D Fluent models to compare parallel and serpentine channels in fuel cell and electrolyzer modes. Implemented patterned wettability in a porous transport layer.

Vapor-Fed PEM Electrolysis | Carnegie Mellon University Sep 2025–Jul 2026

- Ran water collection experiments and analyzed humidity-dependent polarization data to evaluate a coupled COMSOL model at the component level.
- Identified a membrane area ratio of 1.2 that retained 96.8–99.5% of the predicted current density at 1.6 over the tested simulation conditions.

Chip Waste Heat Recovery | Xi'an Jiaotong University Jan 2025–Jun 2025

- Modeled phase change cooling and thermoelectric generation in COMSOL to compare chip temperature stability and power output.

Steel Slag Heat Recovery | Xi'an Jiaotong University Oct 2023–Jul 2024

- Developed a rotating liquid-metal heat exchanger concept and designed a two-loop power generation system for steelmaking waste heat.

Work Experience

Battery Systems Test Intern | Ningbo Deye Technology Jun–Jul 2024

- Ran waterproofing and short-circuit tests, analyzed voltage and current transients, and tested battery management hardware and software.

Patent and Award

Co-inventor Steel slag heat recovery system, Chinese invention patent application CN 121855261 A (2026).

National Second Prize (top 10%) University Student Energy-Saving & Emission Reduction Contest (2024).